

Information Systems and Data Modeling – IT1090



Assignment

Title: Hotel Reservation System for Tourists

Batch Number: 02.02

Group Number: 05

Declaration:

We hold a copy of this assignment that we can produce if the original is lost or damaged.

We hereby certify that no part of this assignment has been copied from any other group's work or from any other source. No part of this assignment has been written / produced for our group by another person except where such collaboration has been authorized by the subject lecturer/tutor concerned.

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Submitted on: 04/06/2023

1.Introduction

With time, technology has evolved and has now integrated itself with every aspect of human life. As a result, life has become efficient and convenient. The Current society prefers products and services that are convenient and accessible.

An online hotel reservation service is considered to be such a service. The aim of this project is to identify the functional and non-functional requirements, as well as to model data, in order to design and build a web application that allows the users to experience the flexible performance and functions that the system provides. This will enable convenience and compatibility with their schedules and other factors, satisfying the requirements of the end users.

While there are many online hotel reservation websites and services in existence, most consist of an outdated and rigid interface where users find it unappealing and hard to navigate. It would be much better if we could address this issue in order to provide an interface that is easier to use and navigate.

An online hotel reservation service will provide customers with the opportunity to make and cancel reservations as well as to provide information about the desired destination of a customer such as; photos of the hotel, amenities available at the hotel, reviews and ratings left by previous customers and much more. This project evaluates the services provided by the suggested system.

2. Hypothetical Scenario

A tourist uses an online hotel reservation system to book lodging for a future trip. They create an account and log in to gain access to the system's capabilities. The user can browse for hotels based on their choices and make reservations by entering the essential information and payment information. The system provides an email confirmation of the booking after confirmation.

The consumer appreciates the hotel services throughout their stay and decides to leave a review and rate their experience on the reviews page. A hotel representative receives a request for a report on guest reservations from the Database Admin. They collect pertinent data from the database and forward the document for decision making and analysis.

If a customer has to cancel their reservation, they can do so through their account. The system validates the request, cancels the reservation, and issues a refund. Meanwhile, the system admin can analyze hotel owner listing requests. They approve the listing after validating the details and then make it available to potential guests.

When the user is ready to check out, they settle any outstanding payments using their preferred payment method. For record-keeping purposes, the system generates an invoice that may be downloaded or sent through email.

3. Requirements Analysis

3.1 Main Requirements for the System

3.1.1 Functional Requirements

The functional requirements outline how the system should respond to user inputs and behave in specific situations. They define the expected system reactions to user actions and determine its behavior in various scenarios.

Guest users, Registered users, Administrator, Hotel representatives, and Database administrator can access the system. Guest users. Registered users and Hotel representatives can access only the front end of the system, while the other users can access both the front and back end of the system. Listed below are the functional requirements of the online hotel reservation system.

User (Guest and Registered)

➤ User requirements

- A Guest user can register into the system.
- A Guest user can search for hotels and listings.
- A Guest user can register and become a Registered user of the system.
- A Registered user should enter the correct details to login to the system.
- A Registered user can also search for hotels and accommodation.
- A Registered user can make reservations.
- A Registered user can cancel reservations.
- A Registered user needs to validate their payment details.
- A Registered user can leave reviews and ratings about a hotel.
- A Registered user can manage/ edit their profile

➤ System requirements

- System needs to store all the Registered user details.
- System allows for the new registrations.
- System should display the listings.
- Check validity of login details.
- System needs to store the payment details and check the validity of the payment details.

- Connect with social media links & display the details.
- System allows customer profile management.

Hotel Representative

➤ User requirements

- A Hotel representative can register into the system.
- A Hotel representative can add a listing for a hotel.
- A Hotel representative can check feedback.
- A Hotel representative can respond to questions posted by customers. (Registered users).
- A Hotel representative can remove a listing.

➤ System requirements

- System needs to store the Hotel representatives.
- System needs to add new listings.
- System needs to notify Hotel representative about questions posted by customers

System Administrator

➤ User requirements

- A System administrator can log into the system.
- A System administrator can update security protocols.
- A System administrator can view reports.
- A System administrator can troubleshoot the system.
- A System administrator should approve a listing.

➤ System requirements

- System needs to generate reports.
- System needs to notify the System administrator about listings.
- System needs to provide the System administrator with access to all functions within the system.

Database Administrator

➤ User requirements

- Database administrator creates query reports.
- Database administrator adds new customers to database.

➤ System requirements

- System generates query reports
- System stores new customers' data in the database.

3.1.2 Non-functional requirements

- **Capacity**
 - System should have the capability to store all the data functions that the customers' need.
 - Store the large number of listings.
- **Availability**
 - The system should be available 24/7.
 - Available for mobile phone app (play store/ apple store)
- **Speed**
 - The customer should be responded to by the system without delay.
 - System does not allow for any delays.
 - Should be accessible to more customers without causing any traffic.
- **Reliability**
 - The system should detect the invalid user credentials if the customer inputs invalid entries into the login.
 - The system should not lose the information entered by the user about their listings.
 - System should correctly validate the payment details entered by the user.
- **Security**
 - Unauthorized access must be restricted by the system and if any security threat is detected, the system should take necessary counter measures.
- **Usability Environment**
 - The website interface should be easy to access for customers without any issues.
- **Serviceability**
 - The system should contain each and every listing accurately.

3.2 Data Requirements for the system:

Personals

- User (Registered and Guest)
- System Administrator
- Hotel Representative

Registered User

- Registered User ID (user_ID)
- Registered User's First Name (user_F_Name)
- Registered User's Last Name (user_L_name)
- Registered User's House Number (user_House_No)
- Registered User's Street (user_Street)
- Registered User's City (user_City)
- Registered User's Country (user_Country)
- Registered User's Postal code (user_Postal_Code)
- Registered User's Date of Birth (user_DOB)
- Registered User's Gender (user_Gender)
- Registered User's Username (user_Username)
- Registered User's Password (user_Password)

Registered User Email

- Registered User ID (user_ID)
- Registered User Email (user_Email)

Registered User Contact No

- Registered User ID (user_ID)
- Registered User Contact No (user_Contact)

System Administrator

- System Administrator ID (admin_ID)
- System Administrator's First Name (admin_F_Name)
- System Administrator's Last Name (admin_L_Name)
- System Administrator's House Number (admin_House_No)
- System Administrator's Street (admin_Street)
- System Administrator's City (admin_City)
- System Administrator's Country (admin_Country)
- System Administrator's postal code (admin_Postal_Code)
- System Administrator's Username (admin_Username)
- System Administrator's Password (admin_Password)

System Administrator Email

- System Administrator ID (admin_ID)
- System Administrator's Email (admin_Email)

System Administrator Contact No

- System Administrator ID (admin_ID)
- System Administrator's Contact Number (admin_Contact)

Hotel Representative:

- Hotel Representative ID (rep_ID)
- Hotel Representative's First Name (rep_F_Name)
- Hotel Representative's Last Name (rep_L_Name)
- Hotel Representative's Hotel ID (hotel_ID) - foreign
- Hotel Representative's House Number (rep_house_No)
- Hotel Representative's City (rep_City)
- Hotel Representative's Country (rep_Country)
- Hotel Representative's Postal code (rep_Postal_Code)
- Hotel Representative's Username (rep_Username)
- Hotel Representative's Password (rep_Password)

Hotel Representative Email:

- Hotel Representative ID (rep_ID)
- Hotel Representative Email (rep_Email)

Hotel Representative Contact No:

- Hotel Representative ID (rep_ID)
- Hotel Representative Contact Number (rep_Contact)

Hotel:

- Hotel ID (hotel_ID)
- Hotel Name (hotel_Name)
- Hotel's Street (hotel_Street)
- Hotel's city (hotel_City)
- Hotel's country (hotel_Country)
- Hotel's Postal code (hotel_Postal_Code)
- Hotel rate (hotel_rate)

Hotel Room type:

- Hotel ID (hotel_ID)
- Hotel Room type (room_type)
- Number of rooms (rooms)

Hotel Email:

- Hotel ID (hotel_ID)
- Hotel Email (hotel_Email)

Hotel Contact:

- Hotel ID (hotel_ID)
- Hotel Contact No (hotel_Contact)

Reservation:

- Reservation ID (reservation_ID)
- Registered User ID (user_ID)
- Hotel's ID (hotel_ID)
- Check-In Date (check_In)
- Check-Out Date (check_Out)
- Reserved room type (room_Type)
- Reserved number of rooms (rooms)

Payment:

- Payment ID (payment_ID)
- Reservation ID (reservation_ID)

- Payment type (payment_Type)
- Payment date (payment_Date)
- Amount paid (amount)

Questions:

- Question ID (answer_ID)
- Registered User's ID (user_ID)
- Hotel's ID (hotel_ID)

Answers:

- Answer ID (answer_ID)
- Question ID (question_ID)
- Hotel Representative ID (rep_ID)

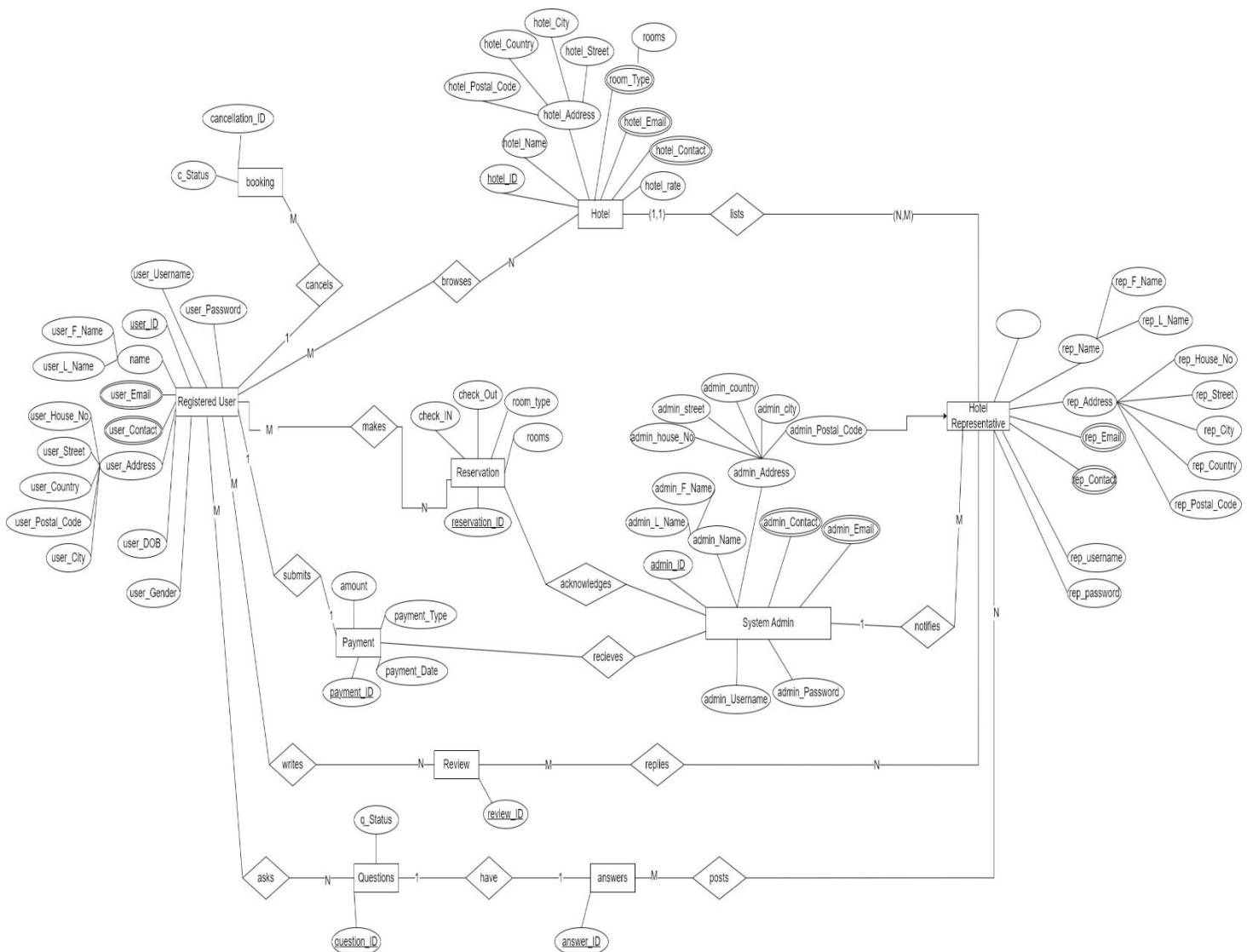
Review:

- Review ID (review_ID)
- Hotel's ID (hotel_ID)
- Registered User's ID (user_ID)

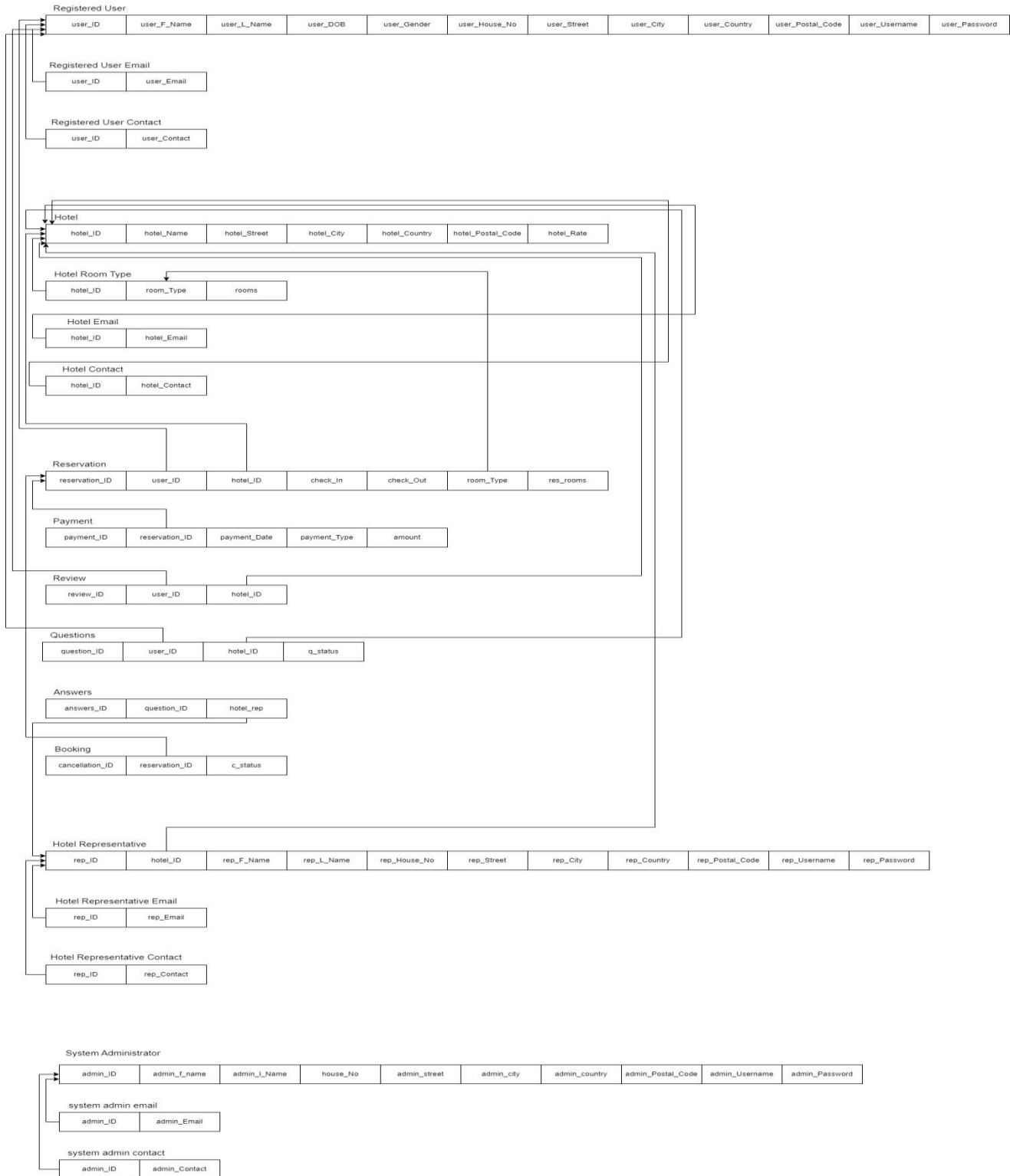
Booking:

- Cancellation ID (cancellation_ID)
- Reservation ID (reservation_ID)
- Cancellation Status (c_Status)

4. ER Diagram



5. Relational Schema



6. SQL Queries to create the database

```
/*-- Registered Users table -- */
CREATE TABLE Registered_Users(
    user_ID CHAR(5),
    user_F_Name VARCHAR(20),
    user_L_name VARCHAR(20),
    user_House_No VARCHAR(20),
    user_Street VARCHAR(20),
    user_City VARCHAR(20),
    user_Country VARCHAR(20),
    user_Postal_code VARCHAR(20),
    user_DOB VARCHAR(15),
    user_Gender CHAR(1),
    user_Username VARCHAR(10),
    user_Password VARCHAR(20),

    CONSTRAINT user_pk PRIMARY KEY (User_ID),
    CONSTRAINT username_unique UNIQUE (user_Username),
    CONSTRAINT chk_user_ID CHECK (User_ID LIKE '[U-u]%),
    CONSTRAINT chk_user_Password CHECK (LEN(user_password) > 8),
    CONSTRAINT chk_user_Gender CHECK (user_Gender LIKE '[mMfF]')
);

/*-- Registered Users Email table -- */
CREATE TABLE Registered_User_Email(
    user_Email VARCHAR(50),
    user_ID CHAR(5),

    CONSTRAINT user_Email_pk PRIMARY KEY (user_ID,user_Email),
    CONSTRAINT user_Email_fk FOREIGN KEY (user_ID) REFERENCES
Registered_Users(user_ID),
    CONSTRAINT chk_user_Email CHECK (user_Email LIKE '%@%')
);

/*-- Registered Users Contact No table -- */
CREATE TABLE Registered_User_Contact(
    user_Contact VARCHAR(20),
    user_ID CHAR(5),

    CONSTRAINT user_Contact_pk PRIMARY KEY (user_ID,user_Contact),
    CONSTRAINT user_Contact_fk FOREIGN KEY (user_ID) REFERENCES
Registered_Users(user_ID),
    CONSTRAINT chk_user_Contact CHECK (LEN(user_Contact) > 0) /* contact number*/
);

/*-- System Administrator table -- */
CREATE TABLE System_Administrators(
    admin_ID CHAR(5),
    admin_F_Name VARCHAR(20),
    admin_L_Name VARCHAR(20),
    admin_House_No VARCHAR(10),
    admin_Street VARCHAR(20),
    admin_City VARCHAR(20),
```

```

        admin_Country VARCHAR(20),
        admin_Postal_Code VARCHAR(10),
        admin_Username VARCHAR(10),
        admin_Password VARCHAR(20),

        CONSTRAINT admin_pk PRIMARY KEY (Admin_ID),
        CONSTRAINT chk_adminID CHECK (Admin_ID LIKE '[A-a]%),
        CONSTRAINT chk_admin_Password CHECK (LEN(admin_password) > 8),
);

/*-- System Administrator Email table -- */
CREATE TABLE System_Administrator_Email(
    admin_Email VARCHAR(50),
    admin_ID CHAR(5),

    CONSTRAINT admin_Email_pk PRIMARY KEY (admin_ID,admin_Email),
    CONSTRAINT admin_Email_fk FOREIGN KEY (admin_ID) REFERENCES
System_Administrators(admin_ID),
    CONSTRAINT chk_admin_Email CHECK (admin_email LIKE '%@%')
);

/*-- System Administrator Contact table -- */
CREATE TABLE System_Administrator_Contact(
    admin_Contact VARCHAR(20),
    admin_ID CHAR(5),

    CONSTRAINT admin_Contact_pk PRIMARY KEY (admin_ID,admin_Contact),
    CONSTRAINT admin_Contact_fk FOREIGN KEY (admin_ID) REFERENCES
System_Administrators(admin_ID),
    CONSTRAINT chk_admin_Contact CHECK (LEN(admin_Contact) > 0)
);

/*-- Hotel table -- */
CREATE TABLE Hotels(
    hotel_ID CHAR(5),
    hotel_name VARCHAR(20),
    hotel_street VARCHAR(20),
    hotel_city VARCHAR(20),
    hotel_country VARCHAR(20),
    hotel_Postal_Code VARCHAR(10),
    hotel_Rate MONEY,

    CONSTRAINT hotel_pk PRIMARY KEY (hotel_ID),
);

/*-- Hotel Rooms table -- */
CREATE TABLE Hotel_Room_Types(
    room_Type VARCHAR(20),
    hotel_ID CHAR(5),
    rooms INTEGER,

    CONSTRAINT room_Type_pk PRIMARY KEY (room_Type,hotel_ID),
    CONSTRAINT room_Type_fk FOREIGN KEY (hotel_ID) REFERENCES Hotels(hotel_ID)
);

/*-- Hotel Email table -- */

```

```

CREATE TABLE Hotel_Email(
    hotel_Email VARCHAR(50),
    hotel_ID CHAR(5),

    CONSTRAINT hotel_Email_pk PRIMARY KEY (hotel_Email,hotel_ID),
    CONSTRAINT chk_hotel_Email CHECK (hotel_email LIKE '%@%'),
    CONSTRAINT hotel_Email_fk FOREIGN KEY (hotel_ID) REFERENCES Hotels(hotel_ID)
);

/*-- Hotel contact table -- */
CREATE TABLE Hotel_Contact(
    hotel_Contact VARCHAR(20),
    hotel_ID CHAR(5)

    CONSTRAINT hotel_Contact_pk PRIMARY KEY (hotel_Contact,hotel_ID),
    CONSTRAINT hotel_Contact_fk FOREIGN KEY (hotel_ID) REFERENCES Hotels,
    CONSTRAINT chk_hotel_Contact CHECK (LEN(hotel_Contact) > 0)
);

/*-- Hotel Representative table -- */
CREATE TABLE Hotel_Representatives(
    rep_ID CHAR(6),
    hotel_ID CHAR(5),
    rep_F_Name VARCHAR(20),
    rep_L_Name VARCHAR(20),
    rep_House_No VARCHAR(10),
    rep_Street VARCHAR(20),
    rep_City VARCHAR(20),
    rep_Country VARCHAR(20),
    rep_Postal_code VARCHAR(10),
    rep_Username VARCHAR(10),
    rep_Password VARCHAR(20),

    CONSTRAINT rep_pk PRIMARY KEY (rep_ID),
    CONSTRAINT rep_fk FOREIGN KEY (hotel_ID) REFERENCES Hotels(hotel_ID),
    CONSTRAINT rep_Username UNIQUE (rep_Username),
    CONSTRAINT chk_rep_Password CHECK (LEN(rep_password) > 8),
);

/*-- Hotel Representative Email table -- */
CREATE TABLE Hotel_Representative_Email(
    rep_Email VARCHAR(50),
    rep_ID CHAR(6),

    CONSTRAINT rep_Email_pk PRIMARY KEY (rep_ID,rep_Email),
    CONSTRAINT rep_Email_fk FOREIGN KEY (rep_ID) REFERENCES
Hotel_Representatives(rep_ID),
    CONSTRAINT chk_rep_Email CHECK (rep_Email LIKE '%@%')
);

/*-- Hotel Representative Contact table -- */
CREATE TABLE Hotel_Representative_Contact(
    rep_Contact VARCHAR(20),
    rep_ID CHAR(6),

    CONSTRAINT rep_Contact_pk PRIMARY KEY (rep_ID,rep_Contact),

```

```

        CONSTRAINT rep_Contact_fk FOREIGN KEY (rep_ID) REFERENCES
Hotel_Representatives(rep_ID),
        CONSTRAINT chk_rep_Contact CHECK (LEN(rep_Contact) > 0)
);

/*-- Reservation table -- */
CREATE TABLE Reservations(
    reservation_ID CHAR(6),
    user_ID CHAR(5),
    hotel_ID CHAR(5),
    check_In VARCHAR(15),
    check_Out VARCHAR(15),
    room_Type VARCHAR(20),
    res_rooms INTEGER,

    CONSTRAINT reservation_pk PRIMARY KEY (reservation_ID),
    CONSTRAINT reservation_fk1 FOREIGN KEY (user_ID) REFERENCES
Registered_Users(user_ID),
    CONSTRAINT reservation_fk2 FOREIGN KEY (hotel_ID) REFERENCES Hotels(hotel_ID),
    CONSTRAINT reservation_fk3 FOREIGN KEY (room_Type,hotel_ID) REFERENCES
Hotel_Room_Types(room_Type,hotel_ID),
    CONSTRAINT chk_reservation_ID CHECK (reservation_ID LIKE '[RS-rs]%),

);

/* -- Cancellation table -- */
CREATE TABLE Cancellations(
    cancellation_ID CHAR(5),
    reservation_ID CHAR(6),
    c_status VARCHAR(10),

    CONSTRAINT cancellation_pk PRIMARY KEY (cancellation_ID),
    CONSTRAINT cancellation_fk FOREIGN KEY (reservation_ID) REFERENCES
Reservations(reservation_ID),
);

/* -- Payment table -- */
CREATE TABLE Payments(
    payment_ID CHAR(5),
    reservation_ID CHAR(6),
    payment_type VARCHAR(15),
    payment_Date VARCHAR(15),
    amount MONEY,

    CONSTRAINT payment_pk PRIMARY KEY (payment_ID),
    CONSTRAINT payment_fk FOREIGN KEY (reservation_ID) REFERENCES
Reservations(reservation_ID),
    CONSTRAINT chk_payment_ID CHECK (payment_ID LIKE '[P-p]%),

);

/* -- Questions table -- */
CREATE TABLE Questions(
    question_ID CHAR(5),
    user_ID CHAR(5),
    hotel_ID CHAR(5),
    q_status VARCHAR (15),

```

```

        CONSTRAINT question_pk PRIMARY KEY (question_ID),
        CONSTRAINT question_fk1 FOREIGN KEY (user_ID) REFERENCES
Registered_Users(user_ID),
        CONSTRAINT question_fk2 FOREIGN KEY (hotel_ID) REFERENCES Hotels(hotel_ID),
        CONSTRAINT chk_question_ID CHECK (question_ID LIKE '[Q-q]%' )
);

/* -- Answers table -- */
CREATE TABLE Answers(
    answer_ID CHAR(7),
    question_ID CHAR(5),
    rep_ID CHAR(6),

    CONSTRAINT answer_pk PRIMARY KEY (answer_ID),
    CONSTRAINT answer_fk1 FOREIGN KEY (rep_ID) REFERENCES
Hotel_Representatives(rep_ID),
    CONSTRAINT answer_fk2 FOREIGN KEY (question_ID) REFERENCES Questions(question_ID),
    CONSTRAINT chk_answer_ID CHECK (answer_ID LIKE 'ANS%')
);

/* --Review table -- */
CREATE TABLE Reviews(
    review_ID CHAR(5),
    user_ID CHAR(5),
    hotel_ID CHAR(5),

    CONSTRAINT review_pk PRIMARY KEY (review_ID),
    CONSTRAINT review_fk1 FOREIGN KEY (user_ID) REFERENCES Registered_Users(user_ID),
    CONSTRAINT review_fk2 FOREIGN KEY (hotel_ID) REFERENCES Hotels(hotel_ID)
);

/*-----Registered Users Details-----*/
SELECT *
FROM Registered_Users;
SELECT *
FROM Registered_User_Email;
SELECT *
FROM Registered_User_Contact;

INSERT INTO Registered_Users
VALUES ('U0001','Emily','Johnson','298','Oak Street','New York','United States',
'10001','12-Mar-1985','F','emily85','qwertg1234'),
('U0002','Daniel','Martinez','135','Maple Avenue','Los Angeles','United
States','90001','05-Jan-1983','M','daniel83','asdjf5678'),
('U0003','Carlos','Gonzalez','072','Elm Street','Madrid','Spain',
'28001','22-May-1964','M','cgonzalez8','zxctv0987'),
('U0004','Sophia','Muller','10','Rosenstrasse','Berlin','Germany',
'10178','02-Nov-1981','f','sophiam81','plmnkb987'),
('U0005','Luca','Rossi','927','Via Roma','Rome','Italy','00185','29-
Jul-1983','M','lucar83','yuio2p765');

INSERT INTO Registered_User_Email
VALUES ('emily85@gmail.com','U0001'),
('daniel83@gmail.com','U0002'),
('cgonzalez8@gmail.com','U0003'),

```

```

        ('sophiam81@gmail.com', 'U0004'),
        ('lucar83@gmail.com', 'U0005');

INSERT INTO Registered_User_Contact
VALUES ('+1 (555) 123-4567', 'U0001'),
       ('+1 (555) 987-6543', 'U0002'),
       ('+34 123456789', 'U0003'),
       ('+49 987654321', 'U0004'),
       ('+39 1234567890', 'U0005');

/*-----System Admin Details-----*/
SELECT *
FROM System_Administrators;
SELECT *
FROM System_Administrator_Email;
SELECT *
FROM System_Administrator_Contact;

INSERT INTO System_Administrators
VALUES ('A0001', 'Michael', 'Anderson', '123', 'Main Street', 'New York', 'United
States', '20001', 'michaela', 'njfh78643'),
       ('A0002', 'Isabella', 'Garcia', '456', 'Palm Avenue', 'London', 'United
Kingdom', 'SW1A 1AA', 'isabellag', 'bshyrf837'),
       ('A0003', 'Liam', 'Martinez', '789', 'Birkenweg', 'Berlin', 'Germany',
'10115', 'liamm', 'urhfb2837'),
       ('A0004', 'Olivia', 'Lefebvre', '321', 'Rue de la Paix', 'Paris', 'France',
'75002', 'olivial', 'aoir87364'),
       ('A0005', 'Aarav', 'Rao', '654', 'MG Road', 'Mumbai', 'India', '400001',
'aaravr', 'amcyr7364');

INSERT INTO System_Administrator_Email
VALUES ('michael.a@wshrs.com', 'A0001'),
       ('isabella.g@wshrs.com', 'A0002'),
       ('liam.m@wshrs.com', 'A0003'),
       ('olivia.l@wshrs.com', 'A0004'),
       ('aarav.r@wshrs.com', 'A0005');

INSERT INTO System_Administrator_Contact
VALUES ('+1 555-123-4567', 'A0001'),
       ('+44 20 1234 5678', 'A0002'),
       ('+49 30 9876 5432', 'A0003'),
       ('+33 1 2345 6789', 'A0004'),
       ('+91 98765 43210', 'A0005');

/*-----Hotel Details-----*/
SELECT *
FROM Hotels;
SELECT *
FROM Hotel_Room_Types;
SELECT *
FROM Hotel_Email;
SELECT *
FROM Hotel_Contact;

INSERT INTO Hotels
VALUES ('H0001', 'Grand Park Hotel', 'Sunset Boulevard', 'Los Angeles', 'United States',
'90001', '99'),

```

```

        ('H0002', 'Royal Palace Hotel', 'Oxford Street', 'London', 'United
Kingdom', 'W1D 1BS', '69'),
        ('H0003', 'Bella Vista Hotel', 'Via Roma', 'Rome', 'Italy', '00185', '75'),
        ('H0004', 'Majestic Hotel', 'Champs-Élysées', 'Paris', 'France', '75008',
'45'),
        ('H0005', 'Golden Sands Resort', 'Bondi Beach Road', 'Sydney', 'Australia',
'2000', '60');

```

```

INSERT INTO Hotel_Room_Types
VALUES ('Standard Room', 'H0001', '5'),
        ('Deluxe Suite', 'H0001', '3'),
        ('Family Suite', 'H0001', '5'),
        ('Penthouse Suite', 'H0001', '1'),
        ('Honeymoon Suite', 'H0001', '3'),

        ('Standard Room', 'H0002', '16'),
        ('Deluxe Suite', 'H0002', '8'),
        ('Family Suite', 'H0002', '4'),
        ('Penthouse Suite', 'H0002', '2'),
        ('Honeymoon Suite', 'H0002', '1'),

        ('Standard Room', 'H0003', '20'),
        ('Deluxe Suite', 'H0003', '10'),
        ('Family Suite', 'H0003', '8'),
        ('Penthouse Suite', 'H0003', '4'),
        ('Honeymoon Suite', 'H0003', '2'),

        ('Standard Room', 'H0004', '15'),
        ('Deluxe Suite', 'H0004', '10'),
        ('Family Suite', 'H0004', '7'),
        ('Penthouse Suite', 'H0004', '5'),
        ('Honeymoon Suite', 'H0004', '3'),

        ('Standard Room', 'H0005', '12'),
        ('Deluxe Suite', 'H0005', '11'),
        ('Family Suite', 'H0005', '6'),
        ('Penthouse Suite', 'H0005', '5'),
        ('Honeymoon Suite', 'H0005', '4');

```

```

INSERT INTO Hotel_Email
VALUES ('info@grandparkhotel.com', 'H0001'),
        ('reservations@royalpalacehotel.com', 'H0002'),
        ('info@bellavistahotel.com', 'H0003'),
        ('reservations@majestichotel.com', 'H0004'),
        ('info@goldensandsresort.com', 'H0005');

```

```

INSERT INTO Hotel_Contact
VALUES ('+1 555-111-1111', 'H0001'),
        ('+44 20 2222 2222', 'H0002'),
        ('+39 06 3333 3333', 'H0003'),
        ('+33 1 4444 4444', 'H0004'),
        ('+61 2 5555 5555', 'H0005');

```

```

/*-----Hotel rep Details-----*/
SELECT *
FROM Hotel_Representatives;

```

```

SELECT *
FROM Hotel_Representative_Email;
SELECT *
FROM Hotel_Representative_Contact;

INSERT INTO Hotel_Representatives
VALUES ('HR0001', 'H0003', 'Emily', 'Johnson', '25', 'Park Avenue', 'New York', 'United
States', '10001', 'emilyj', 'nueyf8734'),
      ('HR0002', 'H0001', 'Oliver', 'Martinez', '10', 'Rue de la Paix', 'Paris',
'France', '75001', 'oliverm', 'uthal8365'),
      ('HR0003', 'H0005', 'Sofia', 'Gonzalez', '7', 'Calle Mayor', 'Madrid',
'Spain', '28013', 'sofiag', '9374bhfyr'),
      ('HR0004', 'H0004', 'Maya', 'Sato', '3-1', 'Shibuya', 'Tokyo', 'Japan',
'150-0002', 'mayas', '029mfu638'),
      ('HR0005', 'H0002', 'Matthew', 'Schmidt', '15', 'Alexanderplatz', 'Berlin',
'Germany', '10178', 'matthews', 'fth64bg80');

INSERT INTO Hotel_Representative_Email
VALUES ('emili.j.hr@wshrs.com', 'HR0001'),
      ('oliver.m.hr@wshrs.com', 'HR0002'),
      ('sofia.g.hr@wshrs.com', 'HR0003'),
      ('maya.s.hr@wshrs.com', 'HR0004'),
      ('matthew.s.hr@wshrs.com', 'HR0005');

INSERT INTO Hotel_Representative_Contact
VALUES ('+1 123-456-7890', 'HR0001'),
      ('+33 987-654-3210', 'HR0002'),
      ('+34 567-890-1234', 'HR0003'),
      ('+81 876-543-2109', 'HR0004'),
      ('+49 234-567-8901', 'HR0005');

/*-----Reservation Table Data-----*/
SELECT *
FROM Reservations;

INSERT INTO Reservations
VALUES ('RS0001', 'U0003', 'H0004', '02/05/2023', '07/05/2023', 'Family Suite', 2),
      ('RS0002', 'U0001', 'H0002', '08/06/2023', '12/06/2023', 'Honeymoon Suite',
1),
      ('RS0003', 'U0005', 'H0003', '10/03/2023', '15/03/2023', 'Standard Room',
5),
      ('RS0004', 'U0002', 'H0001', '05/11/2023', '07/11/2023', 'Standard Room',
3),
      ('RS0005', 'U0004', 'H0005', '16/12/2023', '21/12/2023', 'Deluxe Suite',
1);

/*-----Cancellation Table Data-----*/
SELECT *
FROM Cancellations;

/*-----Payment Table Data-----*/
SELECT *
FROM Payments;

```

```
INSERT INTO Payments
VALUES ('P0001', 'RS0002', 'cash', '25-May-2023', 1188),
      ('P0002', 'RS0005', 'credit', '25-May-2023', 245),
      ('P0003', 'RS0003', 'cash', '25-May-2023', 2250),
      ('P0004', 'RS0001', 'online transfer', '25-May-2023', 405),
      ('P0005', 'RS0004', 'cheque', '25-May-2023', 360);
```

```
/*-----Questions Table Data-----*/
SELECT *
FROM Questions;
```

```
INSERT INTO Questions
VALUES ('Q0001', 'U0004', 'H0005', 'Resolved'),
      ('Q0002', 'U0002', 'H0001', 'Not Resolved'),
      ('Q0003', 'U0005', 'H0003', 'Resolved'),
      ('Q0004', 'U0001', 'H0002', 'Resolved'),
      ('Q0005', 'U0003', 'H0004', 'Not Resolved');
```

```
/*-----Answers Table Data-----*/
SELECT *
FROM Answers;
```

```
INSERT INTO Answers
VALUES ('ANS0001', 'Q0004', 'HR0005'),
      ('ANS0002', 'Q0002', 'HR0001'),
      ('ANS0003', 'Q0005', 'HR0003'),
      ('ANS0004', 'Q0001', 'HR0002'),
      ('ANS0005', 'Q0003', 'HR0004');
```

```
/*-----Reviews Table Data-----*/
SELECT *
FROM Reviews;
```

```
INSERT INTO Reviews
VALUES ('R0001', 'U0004', 'H0005'),
      ('R0002', 'U0002', 'H0001'),
      ('R0003', 'U0005', 'H0003'),
      ('R0004', 'U0001', 'H0002'),
      ('R0005', 'U0003', 'H0004');
```

7. Performance Consideration

(Reference: <https://www.techtarget.com/>)

To perfect our hotel reservation system for all end-users and ensure compatibility with database systems, it is crucial to take performance and security into consideration as they play vital roles in functionality and reliability of the system.

- The user interface should be clear, consistent and visually pleasing.
- Since our system targets global customers, it is essential for the system to be designed to handle the increasing numbers of users, reservations, hotels and representatives in order to achieve the scalability of the system.
- The response time of the system should be minimized to provide a better and convenient user experience.
- Implementing and strengthening the privacy-preserving algorithms in the system followed with redundancy and disaster recovery strategies to ensure continuous availability and reliability.
- In the hotel reservation system, implementation of asynchronous processing is a contributing factor for time consuming tasks such as sending confirmation and validation emails.
- The maintainability and portability of data is vital to consider since the users need to access the system in any place at any time.
- Administrators should be able to update and rectify any errors without delays.
- Enabling users to perform functions such as check-out payments with minimal interference and securely.
- Performance testing should be done regularly to identify and measure response times and validate the performance of the system.

8. Security Requirements

Security is essential when working with sensitive data in a hotel reservation system to safeguard consumer information and keep the system functioning properly.

Adapted security specifications for a hotel reservation system include the following:

- The system administrator and other authorized staff should be the only ones with access to change the system's access permissions.
- Only logged-in users should be able to book hotels (guest users should not be able to do so).
- The system needs to make sure that each email address can only have one account.
- The administrator should be given priority in the system because she is its top authority.
- To preserve secrecy, sensitive data including credit card numbers and personal information should be encrypted.
- The system should use secure techniques to validate financial transactions, such as data encryption and validation.
- Only when login credentials have been verified should full access to the system's features be given.
- Users should be encouraged by the system to make secure passwords for their accounts.

9. Example

100 %
Results Messages

user_ID	user_F_Name	user_L_Name	user_House_No	user_Street	user_City	user_Country	user_Postal_Code	user_DOB	user_Gender	user_Username	user_Password	
1	U0001	Emily	Johnson	298	Oak Street	New York	United States	10001	12-Mar-1985	F	emily85	qwerty1234
2	U0002	Daniel	Martinez	135	Maple Avenue	Los Angeles	United States	90001	05-Jan-1983	M	daniel83	asd95678
3	U0003	Carlos	Gonzalez	072	Elm Street	Madrid	Spain	28001	22-May-1984	M	cgonzale8	zxcv0987
4	U0004	Sophia	Müller	10	Rosenstrasse	Berlin	Germany	10178	02-Nov-1981	F	sophiam21	qwer4567
5	U0005	Luca	Rossi	927	Via Roma	Rome	Italy	00185	29-Jul-1983	M	luca83	yuiop765

user_ID	user_Email	
1	U0001	emily85@gmail.com
2	U0002	daniel83@gmail.com
3	U0003	cgonzale8@gmail.com
4	U0004	sophiam21@gmail.com
5	U0005	luca83@gmail.com

user_ID	user_Contact	
1	U0001	+1 (555) 123-4567
2	U0002	+1 (555) 987-6543
3	U0003	+34 123456789
4	U0004	+49 987654321
5	U0005	+39 1234567890

Query executed successfully. DESKTOP-2HH0U1A\SQLEXPRESS ... DESKTOP-2HH0U1A\DELL (55) Hotel Reservation Syst... 00:00:00 15 rows

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100 %
Results Messages

admin_ID	admin_F_Name	admin_L_Name	admin_House_No	admin_Street	admin_City	admin_Country	admin_Postal_Code	admin_Username	admin_Password	
1	A0001	Michael	Anderson	123	Main Street	New York	United States	20001	michaela	ryh78643
2	A0002	Isabella	Garcia	456	Palm Avenue	London	United Kingdom	SW1A 1AA	isabellag	bahy6337
3	A0003	Liam	Martinez	789	Bismarck	Berlin	Germany	10115	liamm	uinh20337
4	A0004	Olivia	Lefebvre	321	Rue de la Paix	Paris	France	75002	olivial	acv87364
5	A0005	Aarav	Rao	654	MG Road	Mumbai	India	400001	aarav	amqv7364

admin_ID	admin_Email	
1	A0001	michael.a@wshrs.com
2	A0002	isabella.g@wshrs.com
3	A0003	liam.m@wshrs.com
4	A0004	olivia.l@wshrs.com
5	A0005	aarav.r@wshrs.com

admin_ID	admin_Contact	
1	A0001	+1 555-123-4567
2	A0002	+44 20 1234 5678
3	A0003	+49 30 9876 5432
4	A0004	+33 1 2345 6789
5	A0005	+91 98765 43210

Query executed successfully. DESKTOP-2HH0U1A\SQLEXPRESS ... DESKTOP-2HH0U1A\DELL (55) Hotel Reservation Syst... 00:00:00 5 rows

100 %
Results Messages

hotel_ID	hotel_name	hotel_street	hotel_city	hotel_country	hotel_postal_code	hotel_rate	
1	H0001	Grand Park Hotel	Sunset Boulevard	Los Angeles	United States	90001	99.00
2	H0002	Royal Palace Hotel	Oxford Street	London	United Kingdom	W1D 1BS	69.00
3	H0003	Bella Vista Hotel	Via Roma	Rome	Italy	00185	75.00
4	H0004	Majestic Hotel	Champs-Élysées	Paris	France	75008	45.00
5	H0005	Golden Sands Resort	Bondi Beach Road	Sydney	Australia	2000	60.00

hotel_ID	room_Type	rooms
1	H0001	Deluxe Suite 3
2	H0002	Deluxe Suite 8
3	H0003	Deluxe Suite 10
4	H0004	Deluxe Suite 10
5	H0005	Deluxe Suite 11
6	H0001	Family Suite 5
7	H0002	Family Suite 4
8	H0003	Family Suite 8

hotel_ID	hotel_Email	
1	H0003	info@bellavistahotel.com
2	H0005	info@goldensandsresort.com
3	H0001	info@grandparkhotel.com
4	H0004	reservations@majestichotel.com
5	H0002	reservations@royalpalaceh...

hotel_ID	hotel_Contact	
1	H0001	+1 555-111-1111
2	H0004	+33 1 4444 4444
3	H0003	+39 06 3333 33...
4	H0002	+44 20 2222 22...
5	H0005	+61 2 5555 5555

Query executed successfully. DESKTOP-2HH0U1A\SQLEXPRESS ... DESKTOP-2HH0U1A\DELL (55) Hotel Reservation Syst... 00:00:00 40 rows

100 %

Results Messages

	res_ID	hotel_ID	res_F_Name	res_L_Name	res_House_No	res_Street	res_City	res_Country	res_Postal_code	res_Username	res_Password
1	HR0001	H0003	Emily	Johnson	25	Park Avenue	New York	United States	10001	emilj	ruy67734
2	HR0002	H0001	Oliver	Martinez	10	Rue de la Paix	Paris	France	75001	olivem	utha1365
3	HR0003	H0005	Sofia	Gonzalez	7	Calle Mayor	Madrid	Spain	28013	sofiag	9374hMyr
4	HR0004	H0004	Maya	Sato	3-1	Shibuya	Tokyo	Japan	150-0002	mayas	029rnfu138
5	HR0005	H0002	Matthew	Schmidt	15	Alexanderplatz	Berlin	Germany	10178	matthews	th64dy80

res_ID	res_Email
1	HR0001
2	HR0002
3	HR0003
4	HR0004
5	HR0005

res_ID	res_Contact
1	HR0001
2	HR0002
3	HR0003
4	HR0004
5	HR0005

Query executed successfully.

DESKTOP-2HH0U1A\SQLEXPRESS ... DESKTOP-2HH0U1A\DELL (55) Hotel Reservation Syst... 00:00:00 5 rows

100 %

Results Messages

review_ID	user_ID	hotel_ID
1	R0001	U0004
2	R0002	U0002
3	R0003	U0005
4	R0004	U0001
5	R0005	U0003

Query executed successfully. DESKTOP-2HH0U1A\SQLEXPRESS ... DESKTOP-2HH0U1A\DELL (54) Hotel Reservation Syst... 00:00:00 5 rows

100 %

Results Messages

answer_ID	question_ID	res_ID
1	ANS0001	Q0004
2	ANS0002	Q0002
3	ANS0003	Q0005
4	ANS0004	Q0001
5	ANS0005	Q0003

Query executed successfully. DESKTOP-2HH0U1A\SQLEXPRESS ... DESKTOP-2HH0U1A\DELL (54) Hotel Reservation Syst... 00:00:00 5 rows

100 %

Results Messages

question_ID	user_ID	hotel_ID	q_status
1	Q0001	U0004	H0005
2	Q0002	U0002	H0001
3	Q0003	U0005	H0003
4	Q0004	U0001	H0002
5	Q0005	U0003	H0004

Query executed successfully. DESKTOP-2HH0U1A\SQLEXPRESS ... DESKTOP-2HH0U1A\DELL (54) Hotel Reservation Syst... 00:00:00 5 rows

100 %

Results Messages

payment_ID	reservation_ID	payment_type	payment_Date	amount
1	P0001	RS0002	cash	25-May-2023
2	P0002	RS0005	credit	25-May-2023
3	P0003	RS0003	cash	25-May-2023
4	P0004	RS0001	online transfer	25-May-2023
5	P0005	RS0004	cheque	25-May-2023

Query executed successfully. DESKTOP-2HH0U1A\SQLEXPRESS ... DESKTOP-2HH0U1A\DELL (54) Hotel Reservation Syst... 00:00:00 5 rows

100 %

Results Messages

	reservation_ID	user_ID	hotel_ID	check_in	check_out	room_Type	res_rooms
1	RS0001	U0003	H0004	02/05/2023	07/05/2023	Family Suite	2
2	RS0002	U0001	H0002	08/09/2023	12/09/2023	Honeymoon Suite	1
3	RS0003	U0005	H0003	10/03/2023	15/03/2023	Standard Room	5
4	RS0004	U0002	H0001	05/11/2023	07/11/2023	Standard Room	3
5	RS0005	U0004	H0005	16/12/2023	21/12/2023	Deluxe Suite	1

Query executed successfully. DESKTOP-2HHOU1A\SQLEXPRESS ... DESKTOP-2HHOU1A\DELL (54) - Hotel Reservation Syst... 00:00:00 5 rows

Results Messages

cancellation_ID	reservation_ID	c_status
-----------------	----------------	----------

Query executed successfully. DESKTOP-2HHOU1A\SQLEXPRESS ... DESKTOP-2HHOU1A\DELL (54) - Hotel Reservation Syst... 00:00:00 0 rows

#	Student ID	Student Name	Individual Contribution
1	IT22323248	Heidi Hettiarachchi	•Documentation of the Report •Identification of Data Requirements •Drawing the ER-Diagram •Writing Queries to Create and Insert data into Tables •Identification of Performance Consideration
2	IT22338716	Kaavya Raigambandarage	•Identification of Functional Requirements •Identification of Non-Functional Requirements •Identification of Data Requirements •Drawing the ER-Diagram •Writing Queries to Create and Insert data into Tables
3	IT22356222	Tharupathi Bandaranayake	•Documentation of the Report •Identification of Data Requirements •Identification of Security Requirements • Drawing the ER-Diagram • Mapping of Relational Schema
4	IT22350664	Raveesha Abeynayake	•Identification of Functional Requirements •Drawing the ER-Diagram •Mapping in Relational Schema
5	IT22916112	Vishwa Jayampathi Rathnayaka	•Identification of Functional Requirements •Identification of Data Requirements